



YoMusic

Complete Hosting & Deployment Guide

Version 5.0 • YoMusic AI Edition • 2024

■ 100% Free Stack

■ PWA — Mobile +
Desktop

■ AI-Powered Features

■ Deploy in 30 Minutes

Table of Contents

1	Prerequisites — What You Need Before Starting	2
2	Step 1 — Set Up Supabase (Database + Storage)	3
3	Step 2 — Get Your API Keys	4
4	Step 3 — Test Locally on Your Computer	5
5	Step 4 — Deploy Backend to Render.com	7
6	Step 5 — Deploy Frontend to Vercel	9
7	Step 6 — Configure PWA (Mobile Install)	11
8	Testing Checklist — Verify Everything Works	12

9	User Guide — How to Use YoMusic	13
10	AI Features Setup Guide	15
11	Troubleshooting — Common Problems & Fixes	16
12	Cost Breakdown — Everything Free	18

■ Prerequisites — What You Need

Before you start, make sure you have the following installed and ready:

Tool	Version	Download Link	Why Needed
Python	3.10+	python.org/downloads	Runs the backend API
Node.js	18+	nodejs.org	Runs the React frontend
Git	Any	git-scm.com	Push code to GitHub
VS Code	Any	code.visualstudio.com	Edit your code (recommended)
Web Browser	Chrome	Built-in	Testing and admin dashboard

Free Accounts to Create:

Service	URL	Purpose	Free Limit
Supabase	supabase.com	Database + file storage	500MB storage, 2GB/month bandwidth
Render	render.com	Host your Python backend	750 hours/month (enough for 1 app)
Vercel	vercel.com	Host your React frontend	Unlimited hobby projects
Anthropic	console.anthropic.com	AI features (Claude API)	\$5 free credits to start
GitHub	github.com	Store your code	Unlimited public repos

■ TIP: Create all 5 accounts before starting. The whole process takes about 30 minutes once you have everything ready.

■ Step 1 — Set Up Supabase

Supabase is your free database and file storage. This is where all your songs, user accounts, and data live.

1.1 Create Project

1. Go to **supabase.com** → Click **Start your project** → Sign up with GitHub
2. Click **"New Project"** → Choose a name (e.g. **yomusic**) → Set a strong database password → Save it!
3. Select the region **closest to your users** (e.g. West EU for Africa/Europe) → Click **Create project**
4. Wait 2-3 minutes for the project to set up

1.2 Get Your API Keys

These keys connect your backend to the database:

In your Supabase project:
Settings (gear icon) → API

Copy these two values:

Project URL → paste as SUPABASE_URL in your .env file
service_role key → paste as SUPABASE_KEY in your .env file

WARNING: Never share the service_role key publicly!

1.3 Run the Database Schema

This creates all the tables YoMusic needs:

1. In Supabase → click **SQL Editor** (left sidebar)
2. Click **New Query**
3. Open the file **backend/schema.sql** from your YoMusic zip
4. Copy ALL the contents → Paste into the SQL editor
5. Click the green **Run** button
6. You should see: **"Success. No rows returned"**

1.4 Verify Storage Buckets

YoMusic needs two storage buckets for audio files and artwork:

Go to **Storage** in Supabase sidebar → You should see two buckets:

Bucket Name	Purpose	Visibility
music	Stores all audio files (MP3, WAV, etc.)	Public
artwork	Stores cover art and user avatars	Public

■■ If the buckets don't appear: Go to Storage → New Bucket → Name it 'music' → Check 'Public bucket' → Create. Repeat for 'artwork'.

■ Step 2 — Gather All API Keys

You need to set these values in your **backend/.env** file. Copy the template from **backend/.env.example** and fill in your values:

```
# backend/.env

# Supabase (from Step 1)
SUPABASE_URL=https://your-project-id.supabase.co
SUPABASE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...your-service-role-key...

# JWT Secret (make this long and random - min 32 characters)
JWT_SECRET=yomusic-super-secret-key-change-this-now-2024

# Claude AI (from console.anthropic.com → API Keys)
CLAUDE_API_KEY=sk-ant-api03-your-key-here
```

How to Get Your Claude API Key:

1. Go to **console.anthropic.com** → Sign up for free
2. Click **API Keys** in the left sidebar
3. Click **"Create Key"** → Name it "YoMusic"
4. Copy the key (starts with **sk-ant-api03-...**)
5. Paste it as **CLAUDE_API_KEY** in your **.env** file
6. You get **\$5 free credits** to start — enough for thousands of AI requests

■ TIP: The app works perfectly WITHOUT the Claude API key. AI features will gracefully fall back to standard behavior. Add it later when you're ready.

■ Step 3 — Test Locally on Your Computer

Run YoMusic on your own computer first to verify everything works before deploying online.

3.1 Start the Backend

Open a terminal (Command Prompt on Windows, Terminal on Mac/Linux):

```
# 1. Go into the backend folder
cd path/to/musicapp/backend

# 2. Create your .env file from the example
cp .env.example .env
# Then open .env in a text editor and fill in your keys

# 3. Install Python dependencies
pip install -r requirements.txt

# 4. Start the server
python main.py

# You should see:
# INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

■■ Leave this terminal running. Open a NEW terminal for the next step.

3.2 Start the Frontend

```
# Open a NEW terminal window

# 1. Go into the frontend folder
cd path/to/musicapp/frontend

# 2. Create your frontend .env file
cp .env.example .env
# REACT_APP_API_URL=http://localhost:8000 (already set correctly)

# 3. Install Node.js dependencies (takes 2-3 minutes first time)
npm install

# 4. Start the React development server
npm start

# Browser will open automatically at:
# http://localhost:3000
```

3.3 Verify API is Working

Open your browser and go to these URLs to confirm the backend is running:

URL	What You Should See
http://localhost:8000	{"name": "YoMusic API", "version": "2.0.0"}
http://localhost:8000/api/health	{'status': 'ok', 'time': '...'}
http://localhost:8000/api/docs	Interactive API documentation page
http://localhost:3000	YoMusic login page

3.4 First Login — Create Admin Account

1. Open **http://localhost:3000** in your browser
2. Click **Sign up free**
3. Enter your name, email, and password
4. **The very first account created is automatically Super Admin**
5. You will be logged in and see the YoMusic home page
6. Click the **Admin** link in the sidebar to access the dashboard

3.5 Upload Your First Song

1. Click **Upload Music** in the sidebar (admin only)
2. Drag and drop an MP3 or WAV file onto the upload area
3. Fill in Title, Artist, Genre
4. Optionally add cover art (JPG/PNG, 500x500px recommended)
5. Click **Upload Song**
6. Go back to Home — your song should appear!
7. Click the play button to test streaming

■■ Windows Users: If 'pip' is not found, try 'pip3' instead. If 'npm' is not found, restart your computer after installing Node.js.

■ Step 4 — Deploy Backend to Render.com

Render.com hosts your Python backend for free. Users will connect to this URL from anywhere in the world.

4.1 Push Backend to GitHub

```
# In your terminal:

# 1. Go to github.com → New repository → Name: yomusic-backend
# Set to Private → Create repository

# 2. In your backend folder:
cd path/to/musicapp/backend
git init
git add .
git commit -m "Initial YoMusic backend"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/yomusic-backend.git
git push -u origin main

# IMPORTANT: Make sure .env is in .gitignore – never push your keys!
echo '.env' >> .gitignore
```

4.2 Deploy on Render.com

1. Go to **render.com** → Sign up (use GitHub to sign in)
2. Click **New +** → **Web Service**
3. Connect your GitHub account → Select **yomusic-backend** repository
4. Fill in the settings (see table below)
5. Add Environment Variables (click **Add Environment Variable** for each):
6. Click **Create Web Service** → Wait 3-5 minutes for first deploy
7. Copy your Render URL (e.g. **https://yomusic-backend.onrender.com**)

Render Settings:

Setting	Value
Name	yomusic-backend
Environment	Python 3
Region	Frankfurt (EU) or closest to your users

Branch	main
Build Command	pip install -r requirements.txt
Start Command	uvicorn main:app --host 0.0.0.0 --port \$PORT
Plan	Free

Environment Variables to Add on Render:

Variable	Value
SUPABASE_URL	Your Supabase project URL
SUPABASE_KEY	Your Supabase service_role key
JWT_SECRET	Your long random secret string
CLAUDE_API_KEY	Your Claude API key (optional)

■■ FREE PLAN NOTE: Render free tier sleeps after 15 minutes of inactivity. First request after sleep takes ~30 seconds. Upgrade to Starter (\$7/month) for always-on. For testing, free is perfect.

■ Step 5 — Deploy Frontend to Vercel

Vercel hosts your React app for free with a global CDN — your app will load fast from anywhere in the world.

5.1 Push Frontend to GitHub

```
# 1. Go to github.com → New repository → Name: yomusic-frontend
# Set to Private → Create repository

# 2. In your frontend folder:
cd path/to/musicapp/frontend
git init
git add .
git commit -m "Initial YoMusic frontend"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/yomusic-frontend.git
git push -u origin main
```

5.2 Deploy on Vercel

1. Go to **vercel.com** → Sign up (use GitHub to sign in)
2. Click **Add New Project** → Import **yomusic-frontend** repository
3. Framework: Vercel auto-detects **Create React App** — leave as is
4. Click **Environment Variables** → Add one variable:
5. Click **Deploy** → Wait 2-3 minutes
6. Vercel gives you a URL like **https://yomusic.vercel.app**
7. Click the URL to test your live app!

Vercel Environment Variable:

Variable Name	Value
REACT_APP_API_URL	https://yomusic-backend.onrender.com

5.3 Set Up Custom Domain (Optional)

If you have a domain name (e.g. yomusic.com):

```
In Vercel:
Project Settings → Domains → Add Domain → yomusic.com
Follow DNS instructions from Vercel
```

In Render (for backend):

Settings → Custom Domains → Add → `api.yomusic.com`

Update `REACT_APP_API_URL` in Vercel to: `https://api.yomusic.com`

■ TIP: Use Namecheap or Cloudflare to buy a domain. Cloudflare offers free DNS management and the first year of some domains is under \$10.

■ Step 6 — Configure PWA (Mobile Install)

YoMusic is a Progressive Web App — users can install it on their phone like a real app, without any App Store.

How Users Install YoMusic on Their Phone:

Platform	Browser	How to Install
Android	Chrome	Visit the site → Chrome shows 'Add to Home Screen' popup automatically → Tap Install
iPhone	Safari	Visit site in Safari → Tap Share button (box with arrow) → Tap 'Add to Home Screen' → Tap Add
Desktop	Chrome	Visit site → Click install icon in address bar (right side) → Click Install
Desktop	Edge	Visit site → Click install icon → Install

What Users Get After Installing:

- App icon on home screen

Looks and feels like a native app
- Lock screen controls

Play/pause/skip from lock screen
- Instant load

App shell loads from cache instantly
- Full screen (no browser bar)

More space, immersive experience
- Offline cached content

Continues playing if network drops
- Push notifications (future)

Notify users of new music

■ Testing Checklist

Use this checklist to verify every feature works correctly before sharing with users.

Authentication

- Register a new account → first account = Super Admin
- Log out → Log back in
- Register a second account → should be 'user' role
- Admin: go to Admin → Users → change second user to 'premium'

Music Upload (Admin)

- Go to Upload Music in sidebar
- Drag an MP3 file → it should appear in the dropzone
- Fill in title, artist, genre → upload
- Go to Home page → song appears in 'New Releases'
- Try uploading as a regular user → should be blocked (403)

Music Playback

- Click play on any song → audio should start
- Player bar appears at bottom with song info
- Press Space bar → play/pause toggles
- Press Alt+→ → skips to next song in queue
- Click album art in player → full-screen player opens
- Click shuffle → songs play in random order
- Click repeat once → current song repeats

Search

- Click Search → type a song title → results appear
- Switch to AI Search → type 'chill music for evening'
- AI banner shows what it understood → songs appear
- Click a genre pill → songs filtered by genre

Playlists

- Click Library → Create new playlist
- Search for a song → right-click → Add to playlist
- Open playlist → songs appear

- Click AI Name button → AI generates creative name
- Play entire playlist → songs play in order

Premium / Download

- Log in as free user → no download button visible
- Log in as premium user → download button appears in player
- Click download → file downloads to computer
- Try download as free user via API → 403 error returned

AI Features

- Go to AI Mood Player → click 'Workout' preset
- AI generates playlist within 5-10 seconds
- Save playlist to library → appears in Library
- Admin → AI Insights → Generate Insights → report appears

Traffic Monitoring (Admin)

- Admin → Traffic tab → events logged from your browsing
- Admin → Map tab → dots appear for your location
- Admin → Live tab → shows recent activity
- Play several songs → Traffic shows play events

Mobile / PWA

- Open Vercel URL on your phone browser
- App loads and looks correct on small screen
- Hamburger menu (☰) opens sidebar
- Install banner appears → tap Install (Android) or Add to Home Screen (iOS)
- Open installed app → launches full screen
- Play a song → lock screen shows controls

■ User Guide — How to Use YoMusic

For Regular Users (Free Account)

Feature	How to Use	Where to Find
Browse Music	See featured, recent, and recommended songs	Home page
Search	Type in search bar or use AI Search mode	Search page
Play Music	Click play button on any song or double-click row	Anywhere
Like Songs	Click heart icon in player or right-click song	Player bar
Create Playlist	Library → + New Playlist → add songs via right-click	Library page
AI Mood Player	Choose a mood or type what you want to hear	AI Mood Player
History	See recently played songs	Recently Played
Submit Music	Fill out form to request your music be added	Submit Your Music
Install App	Add to Home Screen on mobile for app experience	Browser prompt

For Premium Users

■ Premium users get everything free users have, PLUS the ability to download songs to their device.

To download a song:

- Click any song to play it → Download icon (■) appears in player bar
- OR right-click any song in a list → Download

For Admins

Feature	How to Access	What It Does
Upload Music	Upload Music (sidebar)	Add songs for all users to stream
User Management	Admin → Users tab	View all users, change roles
Promote to Premium	Admin → Users → change role dropdown	Give user download access
Traffic Overview	Admin → Traffic tab	See events, countries, devices

Live Monitoring	Admin → Traffic → Live section	See who's active right now
World Map	Admin → Map tab	See where streams come from
AI Insights	Admin → AI Insights → Generate	Plain English traffic report
Download Log	Admin → Downloads tab	See who downloaded what
Delete Songs	Admin → Songs tab → Delete	Remove any song from platform
Platform Stats	Admin → Overview tab	Total songs, users, plays

■ AI Features Setup Guide

How AI Features Work

All AI features use the Claude API. They are designed to NEVER break your app — if the API key is missing or the AI call fails, every feature gracefully falls back to standard behavior.

Feature	Where	What AI Does	Fallback if No Key
Smart Recommendations	Home page	Reads your last 30 plays, picks matching songs	Shows top played songs
AI Mood Player	AI Mood Player	Generates playlist from mood description	Returns empty, shows error
AI Search	Search page	Understands natural language queries	Falls back to text search
Traffic Insights	Admin dashboard	Writes plain English report from traffic data	Shows basic stats only
AI Playlist Naming	Playlist page	Reads songs, generates creative name	Does nothing

Claude API Usage & Costs

Feature	Tokens Per Call (approx.)	Cost Per Call	Free Credits Cover
Recommendations	~1,500 tokens	~\$0.002	~2,500 calls
Mood Playlist	~2,000 tokens	~\$0.003	~1,600 calls
AI Search	~1,200 tokens	~\$0.0015	~3,300 calls
Traffic Insights	~2,500 tokens	~\$0.003	~1,600 calls
Playlist Naming	~500 tokens	~\$0.0006	~8,000 calls

■ \$5 free credits = thousands of AI calls. For a small platform with 100 users/day, you could run for months on free credits. Top up at console.anthropic.com any time.

■ Troubleshooting — Common Problems

Problem: Backend won't start — 'ModuleNotFoundError'

Cause: You're missing a Python dependency.

Fix:

- `pip install -r requirements.txt`
- If that fails: `pip install fastapi uvicorn python-multipart passlib supabase httpx python-dotenv`

Problem: Frontend shows blank page or 'Cannot GET /'

Cause: The React build hasn't been created yet.

Fix:

- `npm install`
- `npm start` (for development)
- `npm run build` (for production deployment)

Problem: 'CORS error' in browser console

Cause: The backend URL is wrong in your frontend .env file.

Fix:

- Check `REACT_APP_API_URL` in `frontend/.env`
- Make sure it matches your Render URL exactly (no trailing slash)
- After changing `.env`, restart `npm start`

Problem: Songs not playing — audio won't load

Cause: Supabase storage bucket is not set to public.

Fix:

- In Supabase → Storage → music bucket → Settings
- Make sure 'Public bucket' is checked
- Also check the artwork bucket is public

Problem: AI features not working

Cause: Claude API key is missing or incorrect.

Fix:

- Check CLAUDE_API_KEY is set in backend/.env
- Make sure key starts with sk-ant-api03-
- Test: `curl -H "x-api-key: YOUR_KEY" https://api.anthropic.com/v1/models`
- App still works without it — AI falls back gracefully

Problem: Render deploy fails — build error

Cause: Wrong Python version or missing dependency.

Fix:

- Add runtime.txt to backend folder with content: `python-3.11.0`
- Check Render build logs for the exact error
- Make sure requirements.txt is in the backend folder

Problem: Users can't register — 'Email already registered'

Cause: The email exists in database already.

Fix:

- Try a different email
- Or delete the user in Supabase → Table Editor → users table

Problem: Render app is slow on first load (30 sec wait)

Cause: Free tier spins down after 15 min inactivity.

Fix:

- This is normal on free tier
- Upgrade to Render Starter (\$7/mo) for always-on
- Or use a free uptime monitor (e.g. UptimeRobot) to ping every 14 min

Problem: PWA not installable — no install prompt

Cause: Site must be served over HTTPS for PWA to work.

Fix:

- Localhost (http) will not show install prompt — this is normal
- Deploy to Vercel first (uses HTTPS automatically)
- On deployed site, install prompt appears after first visit

Problem: Upload fails — file too large

Cause: Supabase free tier has limits.

Fix:

- Supabase free tier: max 50MB per file
- For larger files, compress your audio to MP3 128kbps
- Or upgrade Supabase storage plan

■ Cost Breakdown — Everything Free

YoMusic is designed to run completely free while you're getting started. Here's exactly what's free and where costs could come in as you grow.

Service	Free Tier	When You'd Pay	Paid Cost
Supabase	500MB storage, 2GB bandwidth/month	More than ~150 songs or many users	\$25/month Pro
Render	750 hrs/month, sleeps after 15 min	Need always-on server	\$7/month Starter
Vercel	Unlimited hobby, 100GB bandwidth	Commercial use / team	\$20/month Pro
Claude AI	\$5 free credits	High AI usage (thousands/day)	Pay per use ~\$3-15/1M tokens
Domain	Free subdomain from Vercel/Render	Custom domain (yomusic.com)	\$10-15/year
GitHub	Unlimited private repos	Larger teams	\$4/month Team

Estimated Monthly Cost by Stage:

Stage	Users	Monthly Cost	Notes
Testing	Just you	\$0	Everything on free tiers
Small launch	1-50 users	\$0	Free tiers still sufficient
Growing	50-500 users	\$7-25	Render + maybe Supabase upgrade
Established	500-5000	\$32-50	Render + Supabase + Vercel Pro
Large platform	5000+	\$100+	Time to talk to investors ■

■ **BOTTOM LINE:** You can run YoMusic with real users for **FREE** until you reach a meaningful scale. Start free, upgrade only when needed.

■ Quick Reference — Important URLs

Resource	URL
----------	-----

Supabase Dashboard	app.supabase.com
Render Dashboard	dashboard.render.com
Vercel Dashboard	vercel.com/dashboard
Claude API Console	console.anthropic.com
API Documentation	https://your-render-url.onrender.com/api/docs
Your Live App	https://your-app.vercel.app
ip-api (Geo)	ip-api.com (free, no key needed, 45 req/min)

■ You're ready to launch YoMusic!

Follow the steps in this guide in order. If you get stuck on anything, the troubleshooting section has fixes for the most common problems. Good luck!